

8.2.21

EE25BTECH11054 - S.Harsha Vardhan

Question 8.2.21 Find the equation of the conic that satisfies the given conditions: vertex (0,0) passing through (5,2) symmetric with respect to Y axis.

Solution: The general conic equation is:

$$g(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1)$$

For a parabola with vertex at the origin and axis along the Y-axis, the focus is $\mathbf{F} = \begin{pmatrix} 0 \\ p \end{pmatrix}$ and the directrix is $y = -p$. This gives $\mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $c = -p$.

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - \mathbf{n} \mathbf{n}^\top = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (2)$$

$$\mathbf{u} = c\mathbf{n} - \mathbf{F} = (-p) \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ p \end{pmatrix} = \begin{pmatrix} 0 \\ -2p \end{pmatrix} \quad (3)$$

$$f = \|\mathbf{F}\|^2 - c^2 = p^2 - (-p)^2 = 0 \quad (4)$$

Substituting these into the general conic equation (1):

$$\mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} 0 & -2p \end{pmatrix}^\top \mathbf{x} + 0 = 0 \quad (5)$$

$$x^2 - 4py = 0 \quad (6)$$

The curve passes through the point (5,2).

$$(5)^2 - 4p(2) = 0 \quad (7)$$

$$25 - 8p = 0 \quad (8)$$

$$p = \frac{25}{8} \quad (9)$$

The final parameters are:

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ -\frac{25}{4} \end{pmatrix}, \quad f = 0 \quad (10)$$

The equation of the parabola is:

$$x^2 = 4 \left(\frac{25}{8} \right) y \quad (11)$$

$$x^2 = \frac{25}{2} y \quad (12)$$

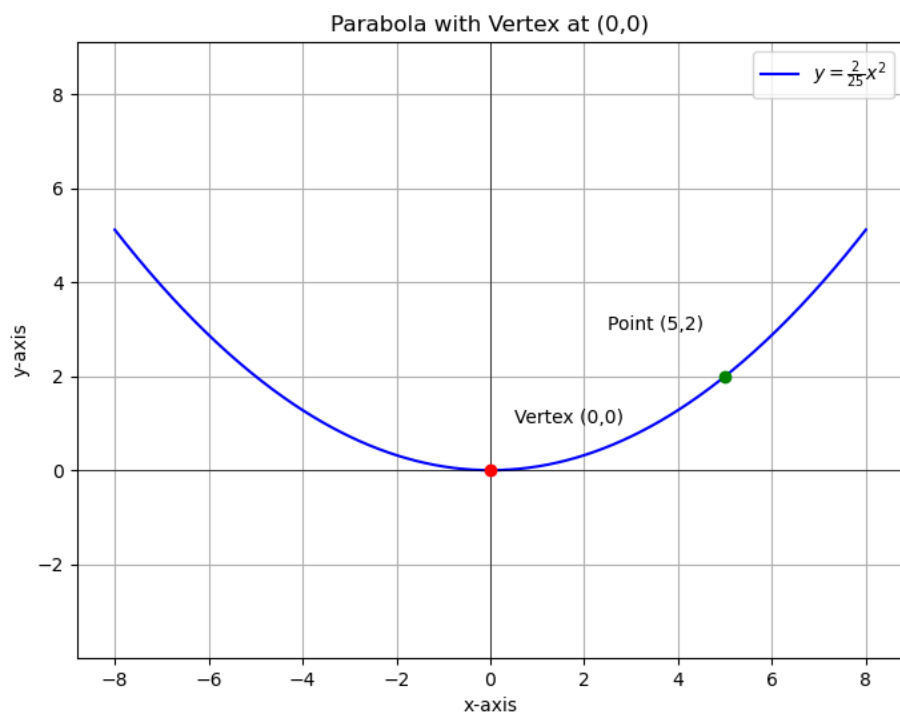


Fig. 0.1